



Name: _____

Teacher: _____

Class: _____

FORT STREET HIGH SCHOOL

2012

PRELIMINARY SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 3: FINAL PRELIMINARY EXAMINATION

Mathematics Extension I

TIME ALLOWED: 2 HOURS

Outcomes Assessed	Questions	Marks
Chooses and applies appropriate arithmetic, algebraic, graphical and geometric techniques to solve problems	1, 2	34
Performs routine arithmetic and algebraic manipulation involving trigonometric identities and applies appropriate trigonometry techniques	3,4	30
Determines the derivative of a function through routine application of the rules of differentiation and relates the derivative of a function to the slope of its graph	5	12
Synthesises mathematical solutions to harder problems and communicates them in appropriate form	6	14

Question	1	2	3	4	5	6	Total	%
Marks	/15	/19	/15	/15	/12	/14	/90	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- **Each new question is to be started in a new booklet**

Question 1: BEGIN A NEW ANSWER BOOKLET

(15 marks)

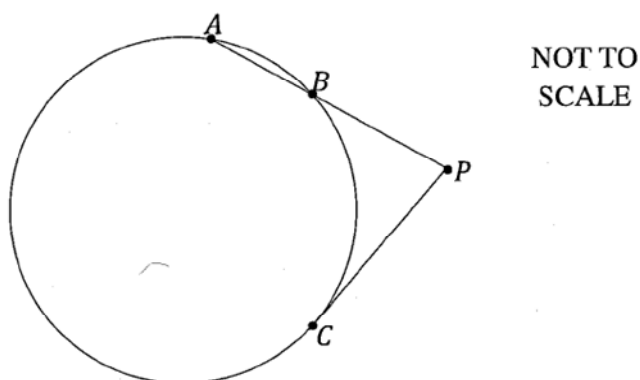
a) Simplify

i. $\frac{1}{\sqrt{(1.2)^2 + (2.1)^2}}$ (correct to 4 significant figures) 1

ii. $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ (in surd form) 1

b) Find the coordinates of the point that internally divides the interval AB in the ratio $3:2$ if A is $(-2,3)$ and B is $(3,-4)$. 2

c) In the diagram the points A, B and C lie on the circle and AB produced meets the tangent from C at the point P .



i. Given that $PC = 12, AB = 7$ and $PB = x$, find x . 2

ii. BC is the diameter of the circle passing through P, B and C . Find the length of BC . 1

d) Find the locus of the point $P(x, y)$ that moves so that P is equidistant from the points $A(3,5)$ and $B(-1,-1)$. 2

e) Solve $|x+1| = 2x-1$ for x . 3

f) Solve $\frac{2t}{1-t} \geq t$ for t . 3

Question 2: BEGIN A NEW ANSWER BOOKLET

(19 marks)

a) For the curve $y = 2 + \sqrt{x-1}$:

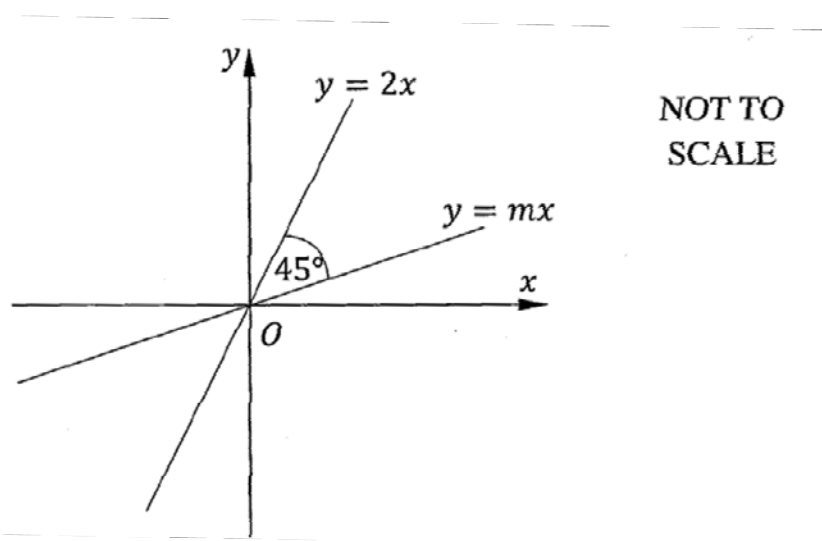
i. Draw a neat $\frac{1}{2}$ page sketch of this curve.

2

ii. State the domain and range for this curve.

2

b) The angle between the lines $y = mx$ and $y = 2x$ is 45° , where $m > 0$, as shown in the diagram below. Find the value of m .

2

c) Find values of a, b and c such that $3x^2 - 7x - 6 \equiv a(x-2)^2 + b(x-2) + c$ for all x .

3

d) The polynomial $P(x) = 2x^3 - 3x^2 + x - 4$ has roots α, β and γ . Find the value of

i. $\alpha + \beta + \gamma$

1

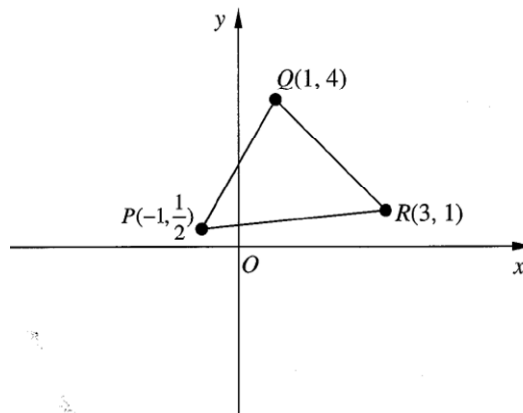
ii. $\alpha^2 + \beta^2 + \gamma^2$

2

e) Find the focus and directrix of the parabola $y = 2x^2 - 4x + 7$.

3

- f) $P\left(-1, \frac{1}{2}\right)$, $Q(1, 4)$ and $R(3, 1)$ are the vertices of a triangle, as shown below.



- iii. Find the gradient of PM , where M is the midpoint of QR . 2
- iv. Show that PM is the perpendicular bisector of QR . 2

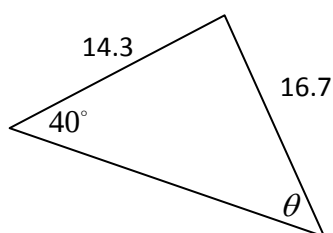
Question 3: BEGIN A NEW ANSWER BOOKLET

(15 marks)

a) Give the exact value for

i. $\sin 60^\circ$ 1

ii. $\tan \frac{5\pi}{6}$ 1

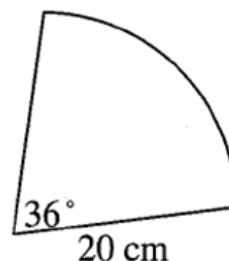
b) Find the value of θ in the following triangle (to the nearest degree). 2c) Draw a neat $\frac{1}{2}$ page sketch of the curve $y = 3\sin\left(2x + \frac{\pi}{3}\right)$ for $0 \leq x \leq 2\pi$ 3

d) For the diagram given:

i. Express 36° in radian measure. 1

ii. Find the exact area of the shape 2

iii. Find the exact perimeter of the shape. 2

e) Two cars leave the point of no return at the same time. Car A averages 80 km/h along a straight road with a bearing of $025^\circ T$ and car B averages 90 km/h along a different straight road with a bearing of $135^\circ T$.

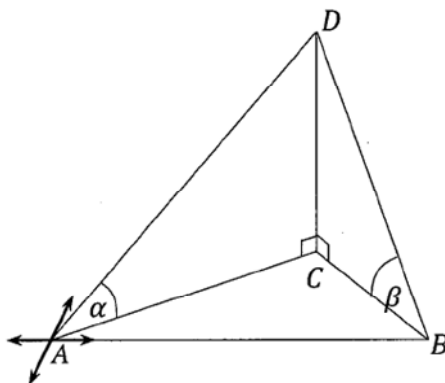
i. Draw a neat sketch showing their positions relative to the point of no return three hours after the cars leave the point of no return. 1

ii. Calculate how far apart the two cars are at this time (to the nearest km). 2

Question 4: BEGIN A NEW ANSWER BOOKLET

(15 marks)

- a) A vertical pole, CD , is positioned so that the angles of elevation of the top of the pole from points A and B on the ground are α and β respectively. The ground is a level horizontal surface and the triangle ABC is right-angled at C . This information is shown in the diagram below.



If point B is due east of point A and point C is on a bearing of $060^\circ T$ from point A ,

show that $\frac{\tan \alpha}{\tan \beta} = \frac{1}{\sqrt{3}}$.

3

b) Prove $\frac{\cot \theta \cos \theta}{\cot \theta + \cos \theta} = \frac{\cos \theta}{1 + \sin \theta}$.

3

c)

i. Write down the angle expansion for $\tan(\theta - \alpha)$.

1

ii. Hence find the exact value of $\tan 165^\circ$.

2

d) Use the transformation for $r \cos(2x + \alpha)$ to express $2\sqrt{3} \cos(2x) - \sin(2x)$ as a single term. State the amplitude, period and phase shift for this expression.

3

e) Find the general solutions for $\cos 2\theta = \sin \theta$.

3

Question 5: BEGIN A NEW ANSWER BOOKLET

(12 marks)

a) Use the rules of differentiation to differentiate with respect to x :

i. $x^2 - 7x + 1$ 1

ii. $(2x^5 - 3)^4$ 2

iii. $(x^4 + 5)(x - 7)^3$ 3

b) Differentiate $f(x) = \frac{1}{x}$ from first principles. 3c) The function $f(x) = x^3 - 3x + 1$ has a root near $x = -2$. Use one iteration of Newton's Method to find a better solution. 3**Question 6: BEGIN A NEW ANSWER BOOKLET**

(14 marks)

a) Express the polynomial $P(x) = x^3 + bx^2 + cx + d$ in its factored form given it has a root at $x = 3$, a y -intercept of 3 and when $P(x)$ is divided by $(x - 2)$ it has a remainder of -3 . 4b) Using the substitution $t = \tan \frac{\theta}{2}$, solve the equation $2 \cos^2 \theta + 3 \cos \theta + 1 = 0$ for $0 \leq \theta \leq 2\pi$ 4c) Find the equation of the normal to the parabola $y = x^2 + 4x - 5$ at the point $(-1, -8)$ and hence find the co-ordinates of the point where it again meets the parabola. 3d) Solve $\cos\left(5x - \frac{\pi}{3}\right) = \sin 3x$ for $0 \leq x \leq 2\pi$. 3

Question 1:

a)

i) 0.4134

ii) $\frac{\sqrt{2}-1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}$

$$= \frac{(\sqrt{2}-1)^2}{2-1}$$

$$= 2 - 2\sqrt{2} + 1$$

$$= 3 - 2\sqrt{2}$$

$$\begin{aligned} \text{b) } x &= \frac{mx_2 + nx_1}{m+n} & y &= \frac{my_2 + ny_1}{m+n} \\ &= \frac{3 \times 3 + 2 \times -2}{3+2} & &= \frac{3 \times -4 + 2 \times 3}{3+2} \\ &= \frac{5}{5} & &= \frac{-6}{5} \\ &= 1 & &= -1\frac{1}{5} \end{aligned}$$

Hence point is $\left(1, -1\frac{1}{5}\right)$

c)

i) $AP \cdot PB = CP^2$

$$x(x+7) = 12^2$$

$$x^2 + 7x = 144$$

$$x^2 + 7x - 144 = 0$$

$$(x+16)(x-9) = 0$$

$$x = -16, 9$$

Reject -16 as x is a length, so x=9.

ii) BC a diameter

$$\Rightarrow \angle BPC = 90^\circ (\angle \text{ in semicircle})$$

Hence:

$$BC^2 = x^2 + 12^2$$

$$= 225$$

$$BC = 15$$

d) For $|PA| = |PB|$ where

$$|PA|^2 = (x-3)^2 + (y-5)^2 \text{ and}$$

$$|PB|^2 = (x+1)^2 + (y+1)^2$$

MARKING

COMMENTS

① answer with
sig figs correct

- Many wrote 0.413 - the 0 before the dp is not significant.

① answer

- Many failed to simplify correctly

① correct subst in
formula

① correct point

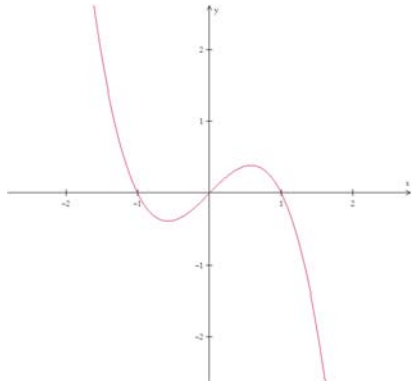
① correct circle
property and
equation

- This formula was poorly known.

① answer

① answer

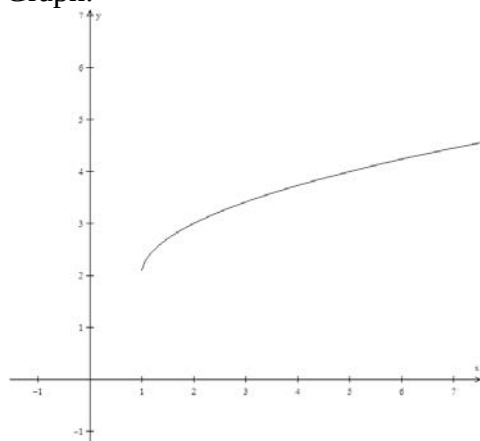
① correct set-up

	MARKING	COMMENTS
$(x-3)^2 + (y-5)^2 = (x+1)^2 + (y+1)^2$ $x^2 - 6x + 9 + y^2 - 10y + 25 = x^2 + 2x + 1 + y^2 + 2y + 1$ $8x + 12y - 32 = 0$ $2x + 3y - 8 = 0$		
e) Two cases: $x+1 = 2x-1 \quad -(x+1) = 2x-1 =$ $2 = x \quad 0 = 3x$ check : check : $LHS = 2+1 \quad LHS = 0+1 $ $= 3 \quad = 1$ $RHS = 4-1 \quad RHS = 0-1$ $= 3 \quad = -1$ $= LHS \quad \neq LHS$ $(accept) \quad (reject)$ Hence $x = 2$ is the only solution.	① correct equation ① cases correct ① checking done	• Full marks given for $8x + 12y - 32 = 0$ but students should realise that any answer is expected to be in simplest form. • Many failed to test cases.
f) $\frac{2t}{1-t} \geq t$ $(1-t)^2 \frac{2t}{(1-t)} \geq t(1-t)^2$ $2t(1-t) \geq t(1-t)^2$ $2t(1-t) - t(1-t)^2 \geq 0$ $(1-t)[2t - t(1-t)] \geq 0$ $(1-t)(2t - t + t^2) \geq 0$ $(1-t)(t + t^2) \geq 0$ $t(1-t)(1+t) \geq 0$  Noting $t \neq 1$ from original expression, hence $t \leq -1$ and $0 \leq t < 1$.	① answer ① method set-up correctly ① justification and/or evidence to support answer ① answer	• Many algebraic errors even when students know the correct method. • Mark deducted of equality sign for $t < 1$.

Question 2:

a) For $y = 2 + \sqrt{x-1}$

i) Graph:

ii) Domain: $x \geq 1$ Range: $y \geq 2$

b) $\tan \theta = \frac{|m_2 - m_1|}{1 + m_1 m_2}$

$$\tan 45 = \frac{|m - 2|}{1 + 2m}$$

$$1 + 2m = |m - 2|$$

Two cases:

$$1 + 2m = m - 2 \quad 1 + 2m = -(m - 2)$$

$$m = -3$$

$$3m = 1$$

reject as $m > 0$

$$m = \frac{1}{3}$$

$$\therefore m = \frac{1}{3}$$

$$\begin{aligned} \text{c) } RHS &= a(x^2 - 4x + 4) + bx - 2b + c \\ &= ax^2 - 4ax + 4a + bx - 2b + c \\ &= ax^2 + (b - 4a)x + 4a - 2b + c \end{aligned}$$

Equating co-efficients:

$$x^2: \quad a = 3$$

$$x: \quad -7 = b - 4a$$

$$b = 5$$

$$\text{const:} \quad -6 = 4a - 2b + c$$

$$c = -8$$

$$\text{i.e. } a = 3, b = 5, c = -8$$

MARKING

COMMENTS

① basic shape & direction correct

① vertex correct

① domain correct

① range correct

① correct use of angle between two lines formula

① correct answer with justification

① correct expansion leading to correct value for a ① b correct① c correct

- Many could not identify and sketch the basic $\sqrt{\quad}$ equation & graph

- This critical point for the curve was often poorly done.

- Well done for those who graphed correctly.

- Many did not use the 2 cases from the absolute value.

- Some did not indicate the condition $m > 0$ and thus get the correct m value.

- Many perfect square expansion errors!

- Many sign errors when calculation the b value

- The constant term often was missing one term.

	MARKING	COMMENTS
d) For $P(x) = 2x^3 - 3x^2 + x - 4$ i) $\alpha + \beta + \gamma = \frac{-b}{a}$ $= \frac{3}{2}$ ii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$ $= \left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right)$ $= \left(\frac{3}{2}\right)^2 - 2\left(\frac{1}{2}\right)$ $= 1\frac{1}{4}$ e) $y = 2x^2 - 4x + 7$ $y = 2(x^2 - 2x + 1 - 1) + 7$ $= 2(x - 1)^2 + 5$ $\frac{1}{2}(y - 5) = (x - 1)^2$ So Vertex is at $(1, 5)$ and $4a = \frac{1}{2}$ $a = \frac{1}{8}$ Hence focus is $\left(1, 5\frac{1}{8}\right)$ and directrix is $y = 4\frac{7}{8}$ f) i) Midpoint is $\left(\frac{1+3}{2}, \frac{4+1}{2}\right)$ or $\left(2, \frac{5}{2}\right)$ Hence gradient of PM: $m = \frac{\frac{5}{2} - 1}{2 - 1}$ $= \frac{2}{3}$ ii) Gradient QR: $m = \frac{4-1}{1-3}$ $= \frac{3}{-2}$ Hence $m_{PM} \times m_{QR} = \frac{2}{3} \times \frac{3}{-2}$ $= -1$	1 answer 1 correct three term expansion for a square 1 answer 1 correct completion of the squares 1 focus correct 1 directrix correct 1 midpoint correct 1 gradient correct 1 shows perpendicular	• Generally well done. • Mostly well done, but some had $-2\alpha\beta\gamma$ instead of the paired roots. • Many simple calculation errors. • The factor of 2 caused many issues with incorrect algebra. • Many used the a value to adjust the x value rather than the y value! • This is a point! • This is an equation (line)! • Generally well done. • Generally well done. • Generally well done.

So as M is the midpoint of QR (from i) and $PM \perp QR$, then PM is the perpendicular bisector of QR (as reqd).

Question 3:

a) Exact values are

i) $\sin 60^\circ$

$$= \frac{\sqrt{3}}{2}$$

ii) $\tan \frac{5\pi}{6}$

$$= -\tan \frac{\pi}{6}$$

$$= -\frac{1}{\sqrt{3}}$$

b) $\frac{\sin \theta}{14.3} = \frac{\sin 40^\circ}{16.7}$

$$\sin \theta = \frac{14.3 \sin 40^\circ}{16.7}$$

$$\theta = \sin^{-1} \left(\frac{14.3 \sin 40^\circ}{16.7} \right)$$

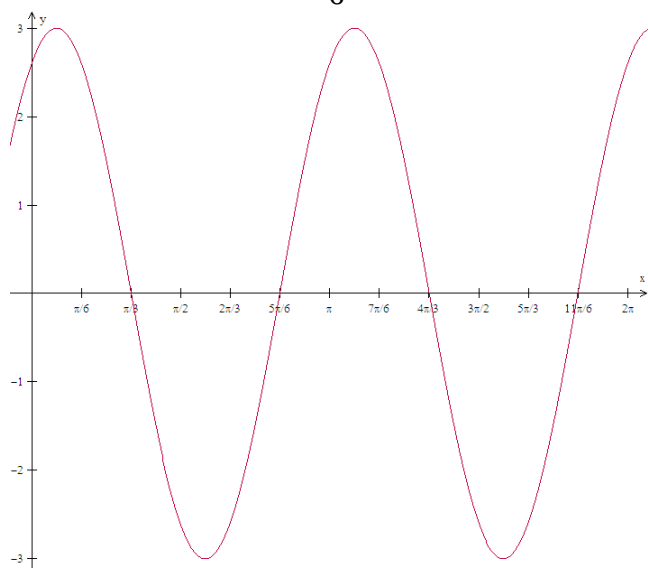
$$= 33.39521023$$

$$\approx 33^\circ \text{ (nearest } ^\circ \text{)}$$

c) $y = 3 \sin \left(2x + \frac{\pi}{3} \right)$: Amplitude is 2, period is

$$\pi \text{ and shifted } 0 = 2x + \frac{\pi}{3} \text{ (i.e. to the left).}$$

$$x = -\frac{\pi}{6}$$

**MARKING**

① conclusion linked correctly

① answer

① answer

① sine rule correct with substitutions

① answer

① amplitude correct

① period correct

① phase shift correct

COMMENTS

- Many either missed or went to great lengths showing the bisection component.

- Some students used composite angle expansion rather than basic trig identities.

- Generally well done.

- Poorly attempted – many did not factor the 2 out to find the correct shift.

- Some stated the correct period, but then did not reflect this correctly in their graph.

- Many need to take greater care graphing.

d)

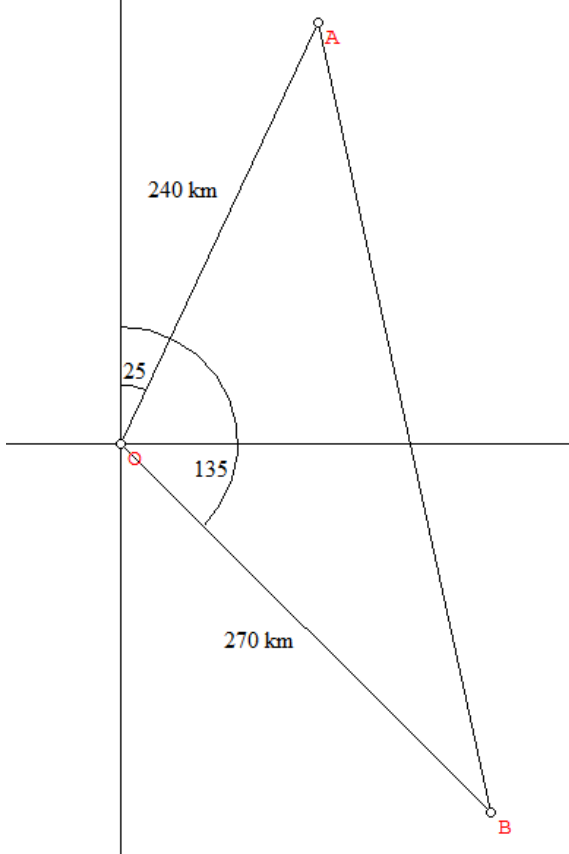
$$\begin{aligned} \text{i) } 36 \times \frac{\pi}{180} \\ = \frac{\pi}{5} \end{aligned}$$

$$\begin{aligned} \text{ii) } A &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} \times 20^2 \times \frac{\pi}{5} \\ &= 40\pi u^2 \end{aligned}$$

$$\begin{aligned} \text{iii) } P &= 2r + r\theta \\ &= 2 \times 20 + 20 \times \frac{\pi}{5} \\ &= 40 + 4\pi \end{aligned}$$

e)

i) Diagram:



$$\begin{aligned} \text{ii) } \angle AOB &= 135 - 25 \\ &= 110 \end{aligned}$$

Hence

$$\begin{aligned} d^2 &= 240^2 + 270^2 - 2 \times 240 \times 270 \times \cos 110^\circ \\ &= 174825.8106 \\ d &= 418.1217652 \\ &\approx 418 \text{ km} \end{aligned}$$

MARKING

COMMENTS

① answer

① correct formula
& substitution

① answer

① answer
① exact value
correct① diagram
correct① cosine rule &
substitutions
correct

① answer

- Many students did not know the radian formulae or could not simplify their answer.

- Many careless errors including calculator in radians, and incorrect distances OA, OB

Question 4:

- a) In $\triangle ABC$ (linking $\triangle$): with $\angle C = 90^\circ$ and $\angle CAB = 90 - 60$ (from bearing), then

$$\tan 30^\circ = \frac{BC}{AC}$$

$$\therefore \frac{BC}{AC} = \frac{1}{\sqrt{3}}$$

In $\triangle ADC$:

$$\tan \alpha = \frac{CD}{AC}$$

In $\triangle ADB$:

$$\tan \beta = \frac{CD}{BC}$$

Hence:

$$\begin{aligned} \frac{\tan \alpha}{\tan \beta} &= \frac{\frac{CD}{AC}}{\frac{CD}{BC}} \\ &= \frac{BC}{AC} \\ &= \frac{1}{\sqrt{3}} \text{ as reqd.} \end{aligned}$$

b)

$$\begin{aligned} LHS &= \frac{\frac{\cos \theta}{\sin \theta} \cdot \cos \theta}{\frac{\cos \theta}{\sin \theta} + \cos \theta} \\ &= \frac{\frac{\cos^2 \theta}{\sin \theta}}{\frac{\cos \theta + \cos \theta \sin \theta}{\sin \theta}} \\ &= \frac{\cos^2 \theta}{\sin \theta} \cdot \frac{\sin \theta}{\cos \theta (1 + \sin \theta)} \\ &= \frac{\cos \theta}{(1 + \sin \theta)} \\ &= RHS \end{aligned}$$

c)

i) $\tan(\theta - \alpha) = \frac{\tan \theta - \tan \alpha}{1 + \tan \theta \tan \alpha}$

MARKING

① set-up for linking triangle

① tan relationships in side triangles

① algebra to reqd solution correct.

① sin/cos substitutions correct

① fraction manipulations correct

① LHS...RHS format correct

① answer

COMMENTS

Students who allocated exact lengths in $\triangle ABC$ were awarded 1 mark.

Mostly well done.

ii)	MARKING	COMMENTS
$\begin{aligned}\tan 165^\circ &= -\tan 15^\circ \\ &= -\tan(60 - 45) \\ &= -\frac{\tan 60 - \tan 45}{1 + \tan 60 \tan 45} \\ &= -\frac{\sqrt{3} - 1}{1 + \sqrt{3}} \left(\times \frac{\sqrt{3} - 1}{\sqrt{3} - 1} \right) \\ &= -\frac{(\sqrt{3} - 1)^2}{3 - 1} \\ &= -\frac{1}{2}(3 - 2\sqrt{3} + 1) \\ &= \sqrt{3} - 2\end{aligned}$	① use of previous part (can use 45 and 30, but resulting fractions are more complex) ① answer	Essential to use tan of difference (the “Hence” in this part); marks were not awarded for using the sum You must rationalize the denominator
d) $r \cos(2x + \alpha) = r \cos 2x \cos \alpha - r \sin 2x \sin \alpha$ With $r \cos \alpha = 2\sqrt{3}$ and $r \sin \alpha = 1$, then $\frac{r \sin \alpha}{r \cos \alpha} = \frac{1}{2\sqrt{3}}$ $\tan \alpha = \frac{1}{2\sqrt{3}}$ $\alpha = \tan^{-1} \frac{1}{2\sqrt{3}}$ $\approx 16.1^\circ$ $r^2 (\sin^2 \alpha + \cos^2 \alpha) = 1^2 + (2\sqrt{3})^2$ $r^2 = 13$ $r = \sqrt{13}$ Hence amplitude $= \sqrt{13}$, period is $\frac{2\pi}{2}$ or π and the phase shift is $\frac{1}{2} \tan^{-1} \left(\frac{1}{2\sqrt{3}} \right)$ which is $\approx 8.05^\circ$ (or 0.14°).	① finds α ① finds r ① interprets & gives values correctly	Phase shift with $2x$ not interpreted well
e) $\cos 2\theta = \sin \theta$ $1 - 2\sin^2 \theta = \sin \theta$ $0 = 2\sin^2 \theta + \sin \theta - 1$ $= 2\sin^2 \theta + 2\sin \theta - \sin \theta - 1$ $= 2\sin \theta (\sin \theta + 1) - (\sin \theta + 1)$ $= (\sin \theta + 1)(2\sin \theta - 1)$	① getting quadratic equation	

Hence:

$$(\sin \theta + 1) = 0$$

$$\sin \theta = -1$$

$$\theta = n\pi + (-1)^n \sin^{-1}(-1)$$

Two cases: odd and even:

Even case: $n = 2m$

$$\theta = 2m\pi + \frac{\pi}{2}$$

$$= (4m+1)\frac{\pi}{2}$$

Odd case: $n = 2m+1$

$$\theta = (2m+1)\pi - \frac{\pi}{2}$$

$$= (4m+2-1)\frac{\pi}{2}$$

$$= (4m+1)\frac{\pi}{2}$$

$$(2\sin \theta - 1) = 0$$

$$2\sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{2}\right)$$

$$= n\pi + (-1)^n \left(\frac{\pi}{6}\right)$$

Two cases: odd and even:

Even case:

$$n = 2m:$$

$$\theta = 2m\pi + \frac{\pi}{6}$$

$$= (12m+1)\frac{\pi}{6}$$

Odd case:

$$n = 2m+1:$$

$$\theta = (2m+1)\pi - \frac{\pi}{6}$$

$$= (12m+5)\frac{\pi}{6}$$

MARKING

COMMENTS

1 answer

1 generator
correctMarks were awarded for
appropriate general
solutions that generated
correct angles.Alternative
Method Marks:

1 set-up correct

Alternative (simpler) method:

$$\cos 2\theta = \cos\left(\frac{\pi}{2} - \theta\right)$$

$$\cos 2\theta = c = \cos\left(\frac{\pi}{2} - \theta\right)$$

Gives 2 equations:

$$2\theta = 2n\pi \pm \cos^{-1} c \quad \frac{\pi}{2} - \theta = \cos^{-1} c$$

Combining:

$$2\theta = 2n\pi \pm \left(\frac{\pi}{2} - \theta\right)$$

Negative case:

$$2\theta = 2n\pi - \left(\frac{\pi}{2} - \theta\right)$$

$$\theta = (4n-1)\frac{\pi}{2}$$

$$\left(\frac{3\pi}{2}, \frac{7\pi}{2}, \frac{11\pi}{2}, \dots\right)$$

Positive case:

$$2\theta = 2n\pi + \frac{\pi}{2} - \theta$$

$$3\theta = (4n+1)\frac{\pi}{2}$$

$$\theta = (4n+1)\frac{\pi}{6}$$

Note: this positive case generates the solutions:

$$\frac{\pi}{6}, \frac{5\pi}{6}, \frac{9\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{21\pi}{6} \dots \text{ which include}$$

the negative case solutions

$$\left(\frac{9\pi}{6} \equiv \frac{3\pi}{2}; \frac{21\pi}{6} \equiv \frac{7\pi}{2}, \text{ etc...}\right) \text{ thus the general}$$

solution is $\theta = (4n+1)\frac{\pi}{6}, n = 0, \pm 1, \pm 2, \dots$

MARKING

COMMENTS

❶ negative case
generator correct

❶ positive case
generator correct

Also Note:

The sin generators from the first method give:

$$(12m+1)\frac{\pi}{6} \text{ gives } \frac{\pi}{6}, \frac{13\pi}{6}, \frac{25\pi}{6}, \dots,$$

$$(12m+5)\frac{\pi}{6} \text{ gives } \frac{5\pi}{6}, \frac{17\pi}{6}, \frac{29\pi}{6}, \dots \text{ and with}$$

the $\theta = (4n+1)\frac{\pi}{2}$ generator common to both methods, these three generate every third term

of the solution $\theta = (4n+1)\frac{\pi}{6}, n = 0, \pm 1, \pm 2, \dots$

MARKING

COMMENTS

Question 5:

a)

i) $2x - 7$

ii) $4(2x^5 - 3)^3 \cdot 10x^4$
 $= 40x^4(2x^5 - 3)^3$

iii) $u = x^4 + 5 \quad v = (x - 7)^3$
 $du = 4x^3 \quad dv = 3(x - 7)^2$

$$\frac{dy}{dx} = u \cdot dv + v \cdot du$$
$$= (x^4 + 5) \cdot 3(x - 7)^2 + (x - 7)^3 \cdot 4x^3$$
$$= (x - 7)^2 [3(x^4 + 5) + 4x^3(x - 7)]$$
$$= (x - 7)^2 (7x^4 - 28x^3 + 15)$$

b) $\frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}$

$$= \lim_{\delta x \rightarrow 0} \frac{\frac{1}{x + \delta x} - \frac{1}{x}}{\delta x}$$

$$= \lim_{\delta x \rightarrow 0} \frac{\frac{x - (x + \delta x)}{x(x + \delta x)}}{\delta x}$$

$$= \lim_{\delta x \rightarrow 0} \frac{x - x - \delta x}{x\delta x(x + \delta x)}$$

$$= \lim_{\delta x \rightarrow 0} \frac{-\delta x}{x\delta x(x + \delta x)}$$

$$= \lim_{\delta x \rightarrow 0} \frac{-1}{x(x + \delta x)}$$

$$= \frac{-1}{x^2}$$

c) $f(-2) = (-2)^3 - 3(-2) + 1$
 $= -1$

$f'(x) = 3x^2 - 3$

$f'(-2) = 3(-2)^2 - 3$
 $= 9$

MARKING

COMMENTS

① answer

① chain rule
correct

① answer

① set-up correct

① product rule
correctly
substituted① answer in
simplest form① first principles
formula with
correct
substitution① algebra
correct to be able
to cancel δx ① limit resolved
correctly① values for
 $x = -2$ correct• This question
generally well done.

Hence

$$\begin{aligned}
 x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\
 &= -2 - \frac{-1}{9} \\
 &= -1\frac{8}{9}
 \end{aligned}$$

Question 6:a) For $P(x) = x^3 + bx^2 + cx + d$ y-intercept of 3 $\Rightarrow d = 3$ root at $x = 3 \Rightarrow P(3) = 0$

$$0 = 27 + 9b + 3c + d$$

$$-30 = 9b + 3c \quad \textcircled{1}$$

Remainder of -3 when division by $(x - 2)$

$$\Rightarrow P(2) = -3$$

$$-3 = 8 + 4b + 2c + d$$

$$-14 = 4b + 2c \quad \textcircled{2}$$

Eliminating c :

$$\textcircled{3} = \textcircled{1} \times 2 : 18b + 6c = -60$$

$$\textcircled{4} = \textcircled{2} \times 3 : 12b + 6c = -42$$

$$\textcircled{3} - \textcircled{4} : 6b = -18$$

$$b = -3$$

In $\textcircled{1}$: $c = -1$

$$\therefore P(x) = x^3 - 3x^2 - x + 3$$

$$\begin{array}{r}
 x^2 - 1 \\
 (x-3) \overline{) x^3 - 3x^2 - x + 3} \\
 \underline{x^3 - 3x^2} \\
 -x + 3 \\
 \underline{-x + 3} \\
 0
 \end{array}$$

Hence

$$P(x) = (x-3)(x-1)(x+1)$$

MARKING

COMMENTS

① formula
correct

① answer

① interprets
given data
correctly to give
initial equations① forms
equations to
solve for b, c
correctly① values for b, c
correct① poly division
or equivalent to
get answer in
factored form

- The main error was getting the function and derivative values in the wrong order.

Many failed to give $P(x)$ in factored form as required.

b) Substituting: $0 = 2\left(\frac{1-t^2}{1+t^2}\right)^2 + 3\left(\frac{1-t^2}{1+t^2}\right) + 1$

$$\begin{aligned} 0 &= 2(1-t^2)^2 + 3(1-t^2)(1+t^2) + (1+t^2)^2 \\ &= 2(1-2t^2+t^4) + 3(1-t^4) + 1+2t^2+t^4 \\ &= 2-4t^2+2t^4+3-3t^4+1+2t^2+t^4 \\ &= 6-2t^2 \end{aligned}$$

$$3 = t^2$$

$$t = \pm\sqrt{3}$$

$$\therefore \tan\left(\frac{\theta}{2}\right) = \pm\sqrt{3}$$

$$\frac{\theta}{2} = \tan^{-1}(\pm\sqrt{3})$$

$$= \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

$$\theta = \frac{2\pi}{3}, \frac{4\pi}{3} \quad (0 \leq \theta \leq 2\pi)$$

Checking $\theta = \pi$ in original equation:

$$LHS = 2\cos^2 \pi + 3\cos \pi + 1$$

$$= 2(-1)^2 + 3(-1) + 1$$

$$= 0$$

$$= RHS$$

$\therefore \theta = \pi$ is also a solution

$$\therefore \theta = \frac{2\pi}{3}, \pi, \frac{4\pi}{3} \quad (0 \leq \theta \leq 2\pi)$$

c) $y' = 2x + 4$

@ $x = -1$: $y' = 2$

So normal has gradient $-\frac{1}{2}$

$$y - 8 = -\frac{1}{2}(x - 1)$$

$$2y + 16 = -x - 1$$

$$x + 2y + 17 = 0$$

is the equation of the normal.

Then, re-arranging:

$$2y = -x - 17$$

And the parabola becomes ($\times 2$):

$$2y = 2x^2 + 8x - 10, \text{ so adding:}$$

$$0 = 2x^2 + 9x + 7$$

$$= 2x^2 + 2x + 7x + 7$$

$$= 2x(x+1) + 7(x+1)$$

MARKING

❶ correct t substitutions

❶ correct t values

❶ correct θ values

❶ Checks and includes $\theta = \pi$ as a solution

❶ equation of normal

COMMENTS

Many students need to learn the ' t ' results; these had to be used as specified in the question.

The $\pm$ was frequently missed.

Some students did not check $\theta = \pi$.

Too many careless errors in this part.

	MARKING	COMMENTS
$0 = (x+1)(2x+7)$ So $x = -1, \frac{-7}{2}$ As $(-1, -8)$ is the original point, the other point has a y value $y = \left(\frac{-7}{2}\right)^2 + 4\left(\frac{-7}{2}\right) - 5$ $= \frac{-27}{4}$ or $\left(-3\frac{1}{2}, -6\frac{3}{4}\right)$ d) $\cos\left(5x - \frac{\pi}{3}\right) = \sin(3x)$ $\cos\left(5x - \frac{\pi}{3}\right) = c = \cos\left(\frac{\pi}{2} - 3x\right)$ $\therefore 5x - \frac{\pi}{3} = 2n\pi \pm \left(\frac{\pi}{2} - 3x\right)$ Positive case: $5x - \frac{\pi}{3} = 2n\pi + \frac{\pi}{2} - 3x$ $8x = 2n\pi + \frac{5\pi}{6}$ $= (12n+5)\frac{\pi}{6}$ $x = (12n+5)\frac{\pi}{48}$ $= \frac{5\pi}{48}, \frac{17\pi}{48}, \frac{29\pi}{48}, \frac{41\pi}{48},$ $\frac{53\pi}{48}, \frac{65\pi}{48}, \frac{77\pi}{48}, \frac{89\pi}{48}$ Negative case: $5x - \frac{\pi}{3} = 2n\pi - \frac{\pi}{2} + 3x$ $2x = 2n\pi - \frac{\pi}{6}$ $= (12n-1)\frac{\pi}{6}$ $x = (12n-1)\frac{\pi}{12}$ $= \frac{11\pi}{12}, \frac{23\pi}{12}$	1 solves equations for x values 1 point correct 1 resolves to cases (or equivalent) 1 solutions from one case (or equivalent) correct 1 solutions from other case (or equivalent) correct	Many failed to realize that $x=1$ had to be one of the solutions – given in the question! Poorly done – too many students did not use general solutions to generate all the solutions in the given domain.